

From Legal Text to Legal Graphs

A Detailed Thesis Presentation on Leakage-Safe Legal Judgment Prediction with
Heterogeneous Graph Neural Networks

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Thesis Presentation

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Roadmap

- 1 Motivation and Research Questions
- 2 Pipeline and Data Construction
- 3 Graph Construction and Leakage Safety
- 4 Graph Analysis and Structural Findings
- 5 Model and Experimental Design
- 6 Prediction Results
- 7 Interpretability and Error Analysis
- 8 Discussion and Conclusion

Thesis at a Glance

Problem

Indian court judgments are long, noisy, multi-hearing, and structurally heterogeneous. Naive text classification risks reading the answer from decision text or identity shortcuts.

Core Idea

Convert judgments into a **leakage-audited heterogeneous legal graph** so the model reasons over cases, parties, arguments, statutes, provisions, precedents, courts, and judges as typed objects.

Data Scale

- ▶ 5 legal domains
- ▶ 75,941 raw hearing documents
- ▶ 73,804 final merged cases
- ▶ 72,735 binary-labelled cases used in the pooled prediction task

Headline Result

Best pooled model:

80.20% accuracy, **0.7959** macro-F1, **0.883**

with a measurable gain over strong text-only references.

Why This Problem Matters

- ▶ **Access to justice** in India is constrained by large judicial backlogs, long litigation cycles, and uneven access to legal assistance.
- ▶ The thesis does **not** argue for automating judges. It argues for **decision support**: tools that organize case material, surface patterns, and assist legal research.
- ▶ Legal Judgment Prediction (LJP) is valuable only if it predicts from **pre-decision signals** such as facts, arguments, statutes, and precedents.
- ▶ If a model can read phrases like “appeal allowed” or memorize judges, the task collapses into **answer recovery**, not legal reasoning.

Thesis Position

A credible legal prediction system must be:

- ▶ structurally faithful,
- ▶ leakage-safe,
- ▶ interpretable,
- ▶ and able to transfer beyond one narrow legal niche.

What Exactly Is Being Predicted?

Target Definition

Outcome prediction from the **appellant/petitioner perspective**:

- ▶ appellant_won (+1)
- ▶ appellant_lost (-1)
- ▶ postponed_or_procedural (0)

Prediction Task Used for the Main Model

The main GNN experiments use the **binary subset**:

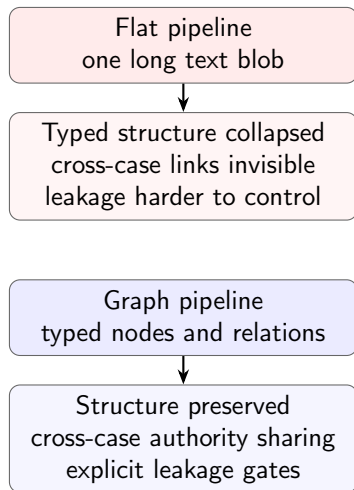
- ▶ win vs. loss only
- ▶ procedural/neutral cases excluded from training
- ▶ outcome-bearing text removed before graph construction

Why This Definition Matters

- ▶ prevents trivial leakage from operative text,
- ▶ makes the task experimentally honest,
- ▶ forces the model to work from pre-decision evidence,
- ▶ aligns evaluation with realistic case-support use.

Why Ordinary Document Classification Is Not Enough

- ▶ Judgments mix **facts**, **procedural history**, **party arguments**, **precedent citations**, and **reasoning** in one long document.
- ▶ Indian judgments are noisy: OCR errors, citation variation, transliterated names, multilingual fragments, inconsistent formatting.
- ▶ One legal matter may produce **multiple hearing-stage documents**; treating each PDF as a separate case corrupts the representation and the label.
- ▶ The core legal signal is **relational**: facts connect to arguments, arguments cite statutes, provisions belong to statutes, cases cite precedents.



Research Questions and Thesis Claims

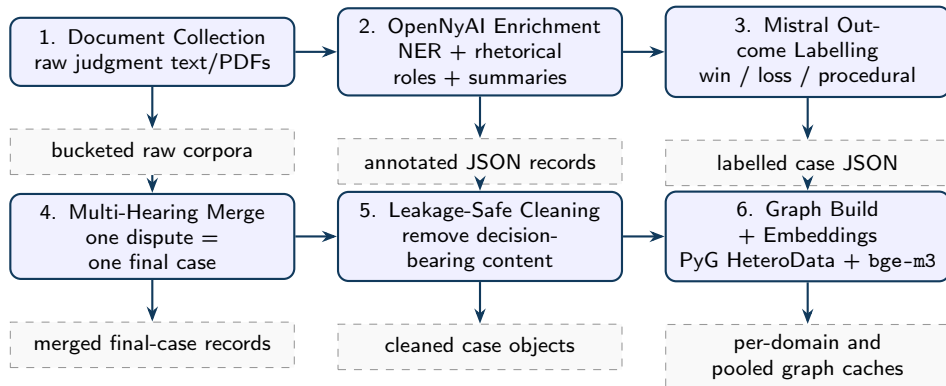
Research Questions

- ▶ **RQ1:** Can Indian court judgments be converted into a legally coherent and leakage-safe structured corpus at scale?
- ▶ **RQ2:** Does a heterogeneous graph representation improve outcome prediction over strong text-only references?
- ▶ **RQ3:** Which parts of the graph actually carry the predictive signal?
- ▶ **RQ4:** Can individual predictions be explained in legally named terms, and does the learned representation transfer to an unseen legal domain?

Claims This Thesis Defends

- ▶ A reasoning-focused heterogeneous graph is a better fit than flat document classification.
- ▶ Leakage can be constrained through schema design, field filtering, rhetorical-role filtering, phrase masking, and temporal gating.
- ▶ The learned signal is primarily legal-text structure plus shared authority nodes, not identity shortcuts.
- ▶ The model is interpretable at both the global and case level.

End-to-End Pipeline

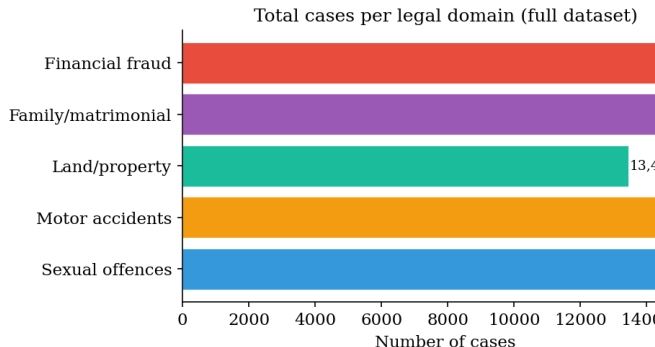


The pipeline is designed so that every expensive stage is cached, auditable, and independently re-runnable.

Corpus Design and Legal Domains

Collection Strategy and Domain Split

- ▶ Source: Indian Kanoon
- ▶ Crawl by **court-month windows** to avoid result truncation
- ▶ Bucket locally by regex/statutory rules instead of relying on search-engine bias
- ▶ Aim: comparable domain scale for fair cross-domain evaluation
- ▶ Domains: Family & Matrimonial, Financial Fraud, Land & Property, Motor Accidents, Sexual Offences



After multi-hearing merging, the domains remain closely balanced in size, which makes pooled evaluation and per-domain comparison meaningful.

OpenNyAI Enrichment Stage

What OpenNyAI Adds to Each Judgment

Component		Output	Why It Matters
Legal NER		statutes, provisions, precedents, courts, judges, parties, lawyers, case numbers	supplies typed legal entities that can become graph nodes
Rhetorical Classifier	Role	sentence-level labels such as preamble, facts, arguments, RPC, RLC	separates usable pre-decision content from outcome-bearing text
Summarizer		structured blocks for preamble, facts, arguments, decision	gives compact sections for outcome labelling and graph text nodes

Why This Is a Good Fit for Indian Legal Text

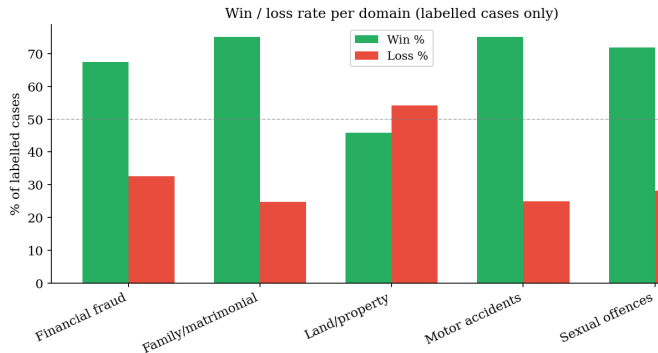
- ▶ It is trained on Indian legal discourse rather than generic text.
- ▶ It recognizes statutes, provisions, precedent citations, and party roles that generic NLP often misses.
- ▶ It produces the typed signals needed both for graph construction and for leakage-safe filtering.

Outcome Target and Class Balance

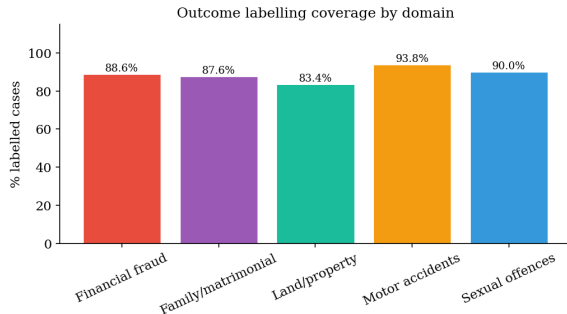
Three Labels from the Appellant's Perspective

- ▶ **won**: relief granted, appeal/petition allowed, conviction quashed, bail granted
- ▶ **lost**: appeal dismissed, relief denied, bail refused
- ▶ **procedural**: remand, adjournment, notice, no clear final outcome

The pooled binary task is roughly **60% win / 40% loss**, so macro-F1 is reported alongside accuracy and ROC-AUC.



Label Quality and Coverage



Coverage is near-complete in every domain; malformed or ambiguous cases are excluded instead of force-labelled.

Cross-Run Reliability Check

A second labelling pass using a different prompt design agreed with the first pass on **84.4%** of 72,795 comparable files.

Bucket	Files	Agreement
Family & Matrimonial	15,017	86.9%
Financial Fraud	14,342	83.8%
Land & Property	13,512	71.4%
Motor Accidents	15,117	89.6%
Sexual Offences	14,807	88.8%
Total	72,795	84.4%

The main disagreement is procedural vs. loss, not clean win vs. loss. That is why the primary prediction task excludes procedural cases.

Multi-Hearing Consolidation

Why Merging Is Necessary

One legal dispute may produce several hearing-stage documents. Without merging:

- ▶ the same dispute becomes multiple case nodes,
- ▶ the final label is not aligned to earlier hearings,
- ▶ shared entities get double-counted as fake independent evidence.

v3 Merge Rule

Merge only when both match:

- ▶ case title, and
- ▶ court registration number.

FIR number is explicitly **not** used, because one FIR can spawn different court proceedings.

Bucket	Inputs	Final Cases
Family & Matrimonial	15,800	15,306
Financial Fraud	15,014	14,615
Land & Property	14,324	14,051
Motor Accidents	15,271	15,236
Sexual Offences	15,532	14,596
Total	75,941	73,804

Merge Outcome

- ▶ 2,111 earlier hearing documents were folded into later final-case records.
- ▶ Sexual-offence matters merge most often; motor-accident cases least.
- ▶ Every merge is logged so the operation remains auditable.

Why a Heterogeneous Graph?

- ▶ A case is not just text. It is a **structured legal object** containing parties, courts, statutes, provisions, precedents, and argument blocks.
- ▶ Different node types play different legal roles. A court is not a statute; a petitioner is not a precedent.
- ▶ A graph preserves **within-case structure** and also **cross-case sharing** through legal authorities cited by many cases.
- ▶ This lets the model aggregate both:
 - ▷ local evidence inside one case, and
 - ▷ relational evidence across similar cases.

Design Principle

Share only what is legally meaningful to share across cases:

- ▶ shared: statutes, provisions, precedents
- ▶ local: parties, lawyers, judges, courts

This keeps the graph useful **without turning identity into a shortcut.**

Reasoning-Focused Graph Schema

17 Node Types in Three Families

- ▶ **Case:** one anchor node per dispute
- ▶ **Text nodes:** Preamble, Facts, Arguments, Petitioner Arguments, Respondent Arguments, Other Lawyer Arguments
- ▶ **Entity nodes:** Petitioner, Respondent, Petitioner Lawyer, Defence Lawyer, Lawyer, Court, Judge, Statute, Provision, Precedent

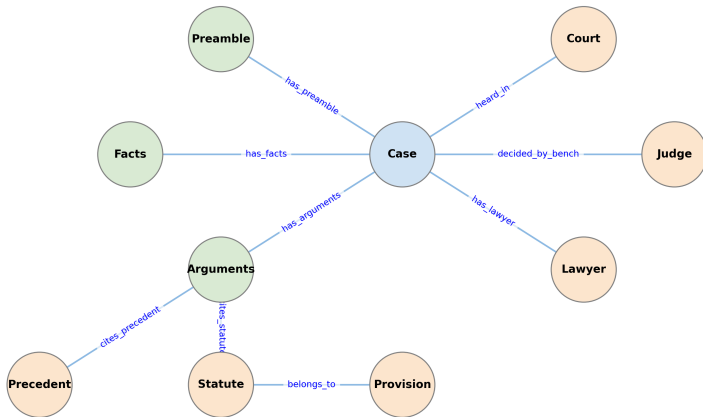
Deliberately Removed

Org, GPE, Date, and Case_Number are excluded from the final reasoning-focused graph because they add noise and possible shortcuts.

Relation Family	Example Edge	Purpose
Case to text	has_facts	attach
Case to entities	heard_in,	encod
	has_petitioner	forum
Argument citations	cites_statute,	prese
	cites_precedent	conte
Provision hierarchy	belongs_to_statute	link s
Argument ownership	party/lawyer to argu- ment blocks	keep cours

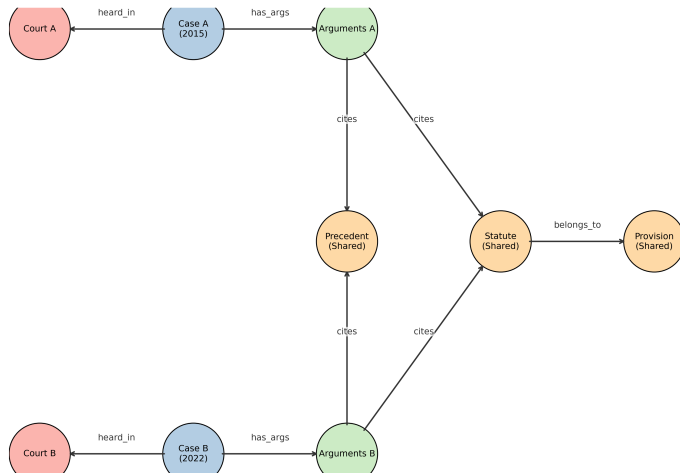
Single-Case Graph Structure

Sample star-graph structure for a single Case node



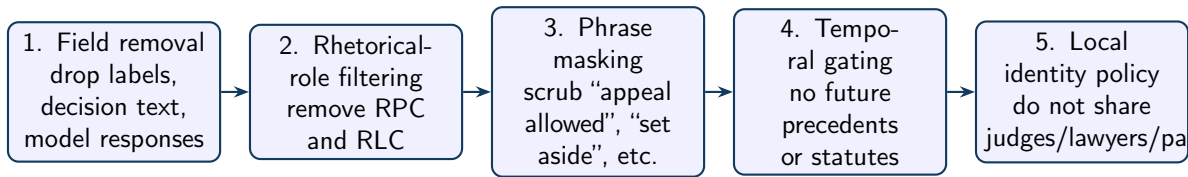
Each case forms a local star centered on the Case node. Text blocks and identity entities remain case-local, while citations from the arguments branch outward to legal authorities.

Cross-Case Sharing and Leakage Control



The key design choice is to share **law** across cases but keep **identity** local. This creates legally meaningful cross-case conduits without allowing judges or litigants to become cheap predictors.

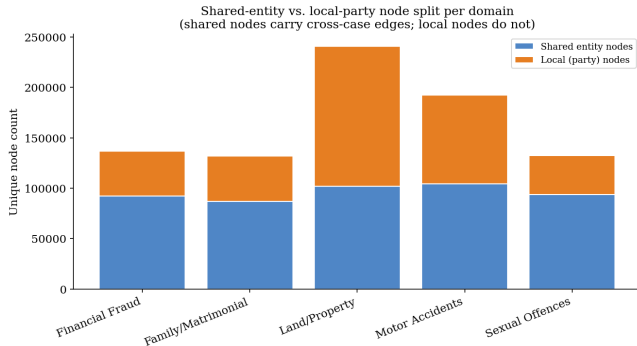
Leakage Prevention Is a First-Class Design Constraint



Interpretation

The model is forced to see a **pre-decision representation of the dispute**: who the parties are, what happened, what each side argued, and what law they cited, but not the ruling itself.

Shared vs. Local Structure



What This Shows

- ▶ Party nodes are overwhelmingly local.
- ▶ Shared nodes are a smaller layer but they carry the graph's cross-case connectivity.
- ▶ Statutes, provisions, and high-frequency precedents form the actual message-passing backbone.

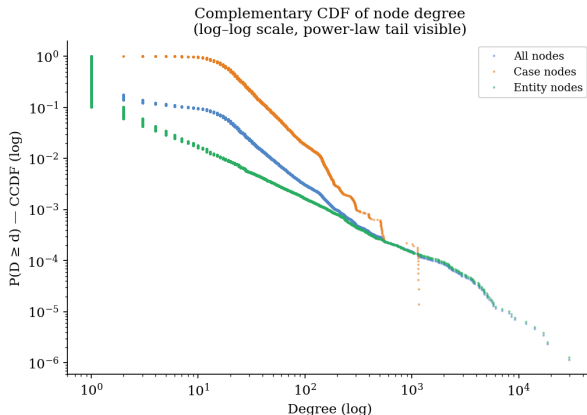
Why It Matters

Cross-case connectivity survives even after the thesis intentionally prevents identity-proxy sharing.

Hub Dominance and Heavy-Tailed Structure

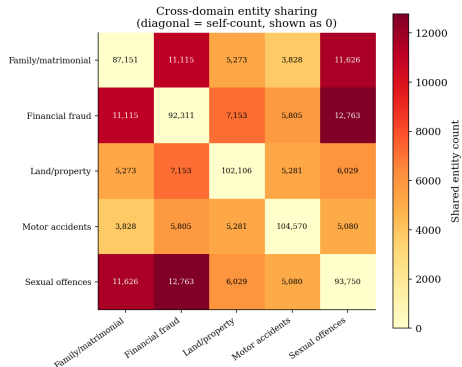
Hub Node	Type	Degree
IPC	Statute	29,602
CR.P.C.	Statute	18,704
Supreme Court	Court	17,294
Apex Court	Court	14,242
Indian Penal Code	Statute	12,009
Section 482	Provision	8,732
Section 420	Provision	6,285
Constitution of India	Statute	5,950

Every top hub bridges all five legal domains.
These are the nodes through which
cross-case signal can actually travel.



The degree distribution is heavy-tailed: most nodes are leaves or near-leaves, while a small number of legal authorities act as super-hubs.

How the Legal Domains Connect

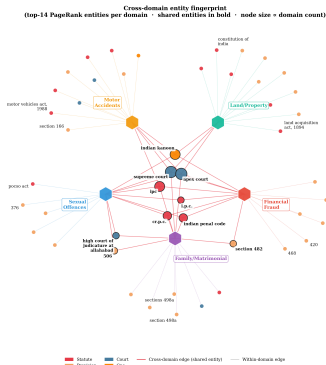


Main Pattern

- ▶ Family, financial fraud, and sexual offences share heavily because they are IPC/CrPC-driven.
- ▶ Motor accidents shares moderately through procedural authorities and appellate courts.
- ▶ Land & property is more weakly connected because it has a different statutory backbone.

The domains are related enough for pooled training to matter, but different enough that transfer remains non-trivial.

Domain Fingerprints in the Graph



Three-Tier Reading

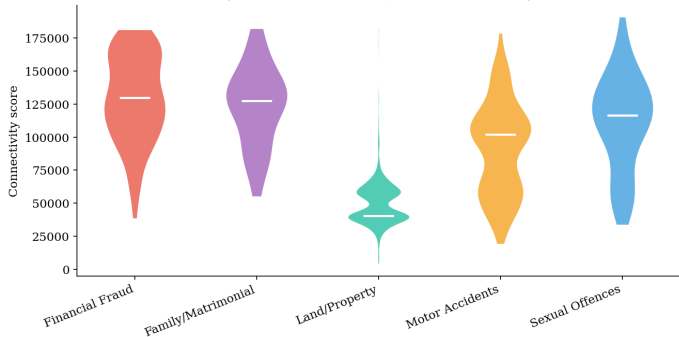
Universal bridges: Supreme Court, Apex Court.

Criminal-procedure bridges: IPC, CrPC, Section 482.

Domain anchors: Motor Vehicles Act, POCSCO, Land Acquisition Act.

Connectivity Predicts Modelling Difficulty

Case connectivity score distribution per domain
(violin = full distribution, white line = median)



Reading the Figure

- ▶ High-connectivity cases cite more shared legal authorities and can absorb richer neighbourhood signal.
- ▶ Land/property and financial fraud have broader, higher-median connectivity distributions.
- ▶ Motor accidents is structurally narrow: many cases cite the same limited authority set.

Low-connectivity cases are intrinsically harder because they have less useful shared context to aggregate.

Why HGT Instead of a Simpler GNN?

- ▶ The graph has **many node types** with different feature semantics.
- ▶ It also has **many relation types**:
`cites_statute`, `heard_in`,
`belongs_to_statute`, `has_petitioner`, and more.
- ▶ HGT gives:
 - ▷ per-node-type projections,
 - ▷ per-relation message functions,
 - ▷ attention over typed neighbourhoods.
- ▶ That makes it better suited than homogeneous GNNs or simple relation-shared models.

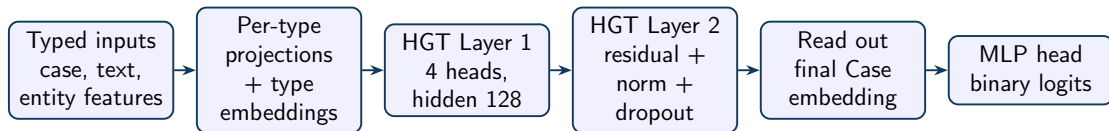
Architectural Intuition

A legal graph should not force statutes, arguments, parties, and courts into one shared geometry before message passing begins.

Two-Layer Choice

Two HGT layers are enough to move information between a case and the shared legal authorities cited in its argument structure.

Model Architecture



Training

AdamW, learning rate 10^{-3} , weight decay 10^{-5} , 60 epochs, early stopping by validation macro-F1, class-balanced cross-entropy.

Node Features

bge-m3 text embeddings for text nodes, canonical-name embeddings plus scalar features for entity nodes, bucket metadata for case nodes.

Experimental Protocol

Evaluation Regime

- ▶ **Transductive**: the graph structure is visible, but test labels are masked.
- ▶ **5-fold cross-validation**: seeds 42–46.
- ▶ Split per fold: 72% train, 8% validation, 20% test.

Settings

- ▶ 1 pooled cross-bucket setting
- ▶ 5 per-domain settings
- ▶ later: 1 zero-shot unseen-domain setting

Metrics

- ▶ Primary: macro-F1
- ▶ Secondary: accuracy, ROC-AUC
- ▶ Supplementary: per-class precision/recall/F1 in logs

Ablation Logic

Each ablation removes one structural or featural assumption: `text_only`, `no_cross_case`, `no_names`, `hierarchical_enc`, `section_sep_enc`, `case_node_minimised`, and `party_args_lr_decay`.

Reference Baselines

Configuration	Setting	Accuracy	Macro-F1
Majority baseline	fold-0 reference	0.609	0.378
Raw-feature logistic regression	fold-0 reference	0.744	0.736
HGT baseline	5-fold pooled mean	0.7811 ± 0.0055	0.7759 ± 0.0043

Interpretation

A strong non-graph reference already reaches 0.744 accuracy, so this is not a trivial task. The heterogeneous graph is justified only if it improves on that baseline without relying on leakage, which the main tables show that it does.

Main Pooled Cross-Bucket Results

Configuration	Accuracy	Macro-F1	ROC-AUC
HGT baseline	0.7811 ± 0.0055	0.7759 ± 0.0043	0.8674
+ section_sep_enc	0.7892 ± 0.0093	0.7831 ± 0.0086	0.8712
+ no_names	0.7866 ± 0.0072	0.7797 ± 0.0066	0.8665
+ no_cross_case	0.7837 ± 0.0047	0.7767 ± 0.0032	0.8658
+ hierarchical_enc	0.7811 ± 0.0033	0.7754 ± 0.0026	0.8655
+ text_only	0.7732 ± 0.0063	0.7671 ± 0.0052	0.8561
+ case_node_minimised	0.7755 ± 0.0030	0.7697 ± 0.0023	0.8560
+ party_args_lr_decay	0.8020 ± 0.0036	0.7959 ± 0.0028	0.8831

Best pooled result on $N = 72,735$ cases: `party_args_lr_decay` is the strongest overall configuration.

What the Ablations Actually Say

Five Concrete Findings

- ▶ **Text dominates:** the `text_only` graph remains surprisingly strong, so the core signal is still carried by legal text representations.
- ▶ **Identity is removable:** `no_names` is at or slightly above baseline, which directly argues against party/judge name leakage as the main source of performance.
- ▶ **Cross-case sharing helps, but less than expected:** removing shared authorities is close to baseline in the pooled transductive regime.
- ▶ **Better text organization helps selectively:** `section_sep_enc` and `hierarchical_enc` help more in long-argument domains.
- ▶ **Best pooled model stabilizes the argument stream:** `party_args_lr_decay` improves by about +2.1 accuracy points and +2.0 macro-F1 points over the baseline.

Interpretive Summary

The graph's benefit is not “pure hub traversal.” It is better understood as **structured aggregation of legal context** layered on top of already strong text features.

Per-Domain Results

Bucket	Accuracy	Macro-F1	ROC-AUC
Family & Matrimonial	0.8217 ± 0.0079	0.8042 ± 0.0088	0.89
Financial Fraud	0.7683 ± 0.0093	0.7577 ± 0.0103	0.84
Land & Property	0.8083 ± 0.0050	0.7916 ± 0.0060	0.86
Motor Accidents	0.8420 ± 0.0043	0.8147 ± 0.0052	0.90
Sexual Offences	0.7752 ± 0.0071	0.7538 ± 0.0067	0.84
Cross-bucket pooled	0.7811 ± 0.0055	0.7759 ± 0.0043	0.86

Stable Difficulty Ordering

- ▶ easiest: **Motor Accidents**
- ▶ then: Land & Property and Family & Matrimonial
- ▶ hardest: Financial Fraud and Sexual Offences

Why?

Motor-accident cases sit on repetitive, highly connected legal backbones. Fraud and sexual-offence cases are more ambiguous, more structurally diverse, and noisier at the outcome-label level.

Zero-Shot Transfer to a Fully Unseen Legal Domain

Setting	N	Accuracy	Macro F1	ROC AUC
Majority baseline (food safety)	11,769	0.543		
Cross-bucket $\rightarrow$ food safety	11,769	0.6079 ± 0.0054	0.6034	
In-domain pooled reference	72,735	0.8020 ± 0.0036	0.7959	

What This Means

- ▶ No fine-tuning was used.
- ▶ Accuracy is **6.5 points above** the majority baseline.
- ▶ The model clearly loses substantial performance out of domain.
- ▶ But the transfer is non-trivial and stable across folds.

Interpretation

The pooled model learns a **partially domain-general legal prior**, not just a memorized list of the training buckets' statutes.

Four Levels of Interpretation

- ▶ **Representation level:** what changed after graph message passing?
- ▶ **Component level:** which node and edge types matter?
- ▶ **Error level:** where does the model fail?
- ▶ **Case level:** why did one prediction happen?

Case-Level Pipeline

Frozen inference → FAISS retrieval index → heterogeneous PGExplainer → ranked legal evidence → LLM verbalisation.

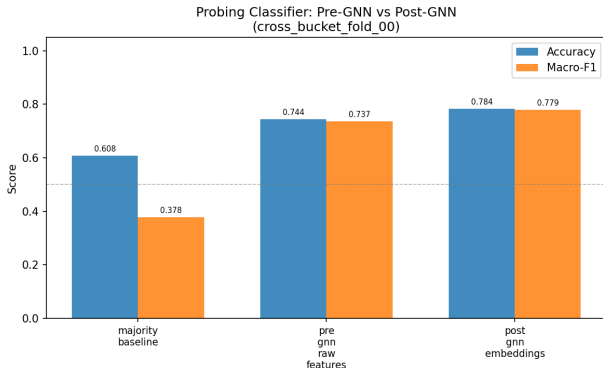
Phase	Script	Output
1–2	inference + index	embeddings, FAISS index
3	explainer training	hetero PGExplainer
4	importance extraction	ranked statutes, precedents, arguments
5	LLM translation	grounded natural language explanation

What Did the GNN Learn Beyond the Raw Features?

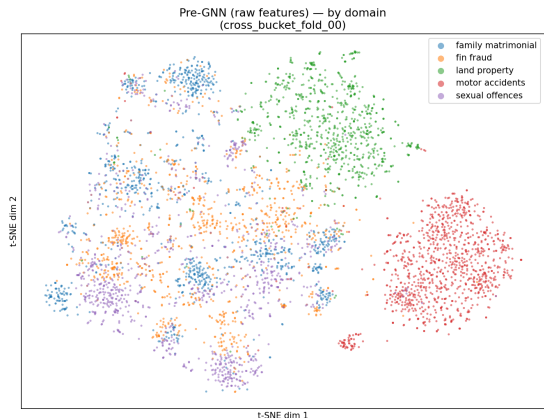
Probe Input	Acc.	Macro-F1
Majority baseline	0.6083	0.3782
Pre-GNN raw features	0.7443	0.7372
Post-GNN embedding	0.7840	0.7792

Key Point

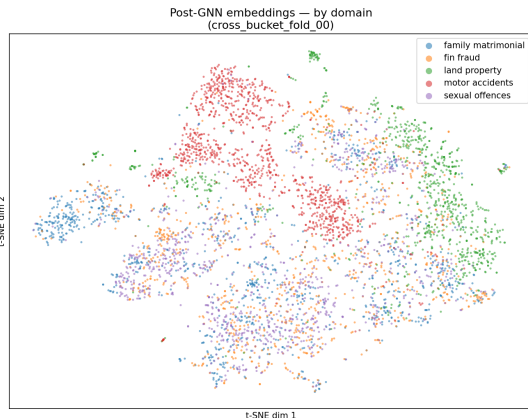
The post-GNN representation is **16×** smaller than the pre-GNN raw feature vector, yet it is almost **4 points more accurate**. The gain comes from message passing, not probe capacity.



Embedding Geometry: Domain Structure



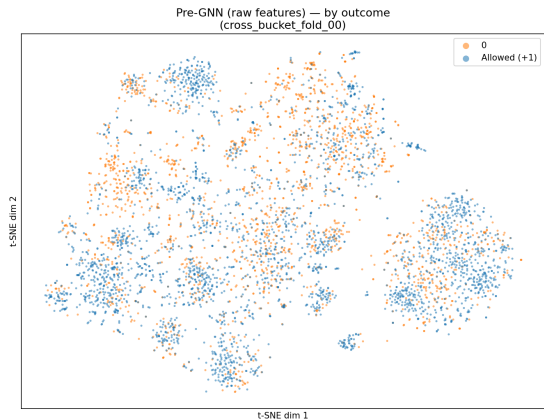
Pre-GNN embeddings colored by domain



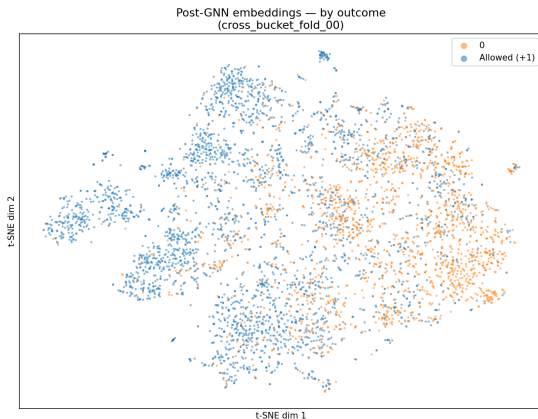
Post-GNN embeddings colored by domain

Domain structure is already present before message passing, which shows that raw legal text is informative about legal area.

Embedding Geometry: Outcome Structure



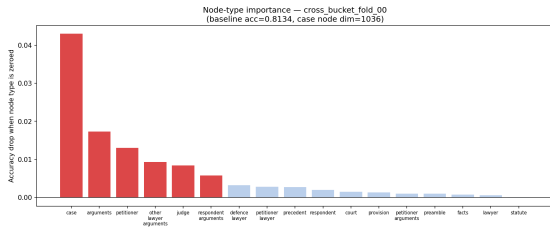
Pre-GNN embeddings colored by outcome



Post-GNN embeddings colored by outcome

After message passing, outcome structure becomes much more visible **within** domain neighbourhoods.
This is the geometric view of the graph's contribution.

Node-Type Importance



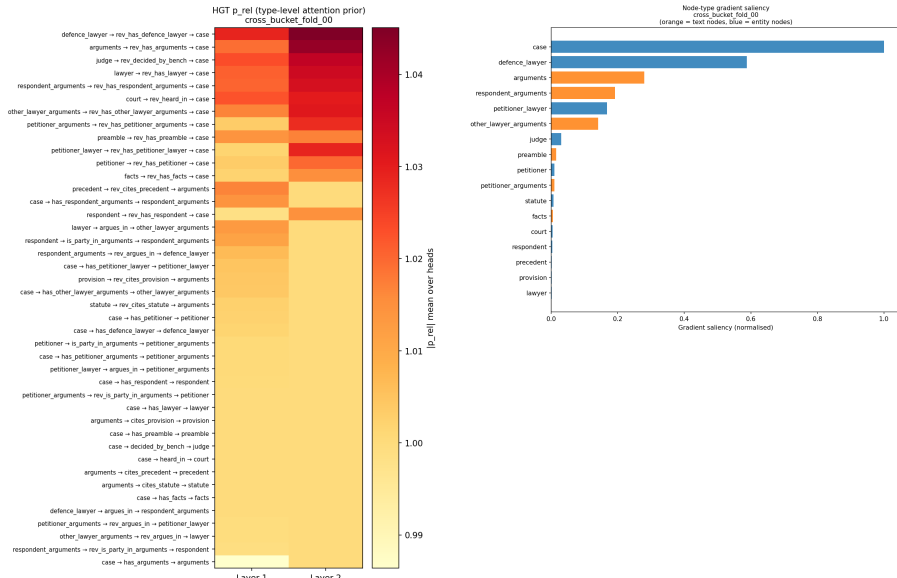
Zeroing Result

The Case node matters most, followed by the argument-bearing text stream and immediate participants such as petitioners and judges.

Interpretation

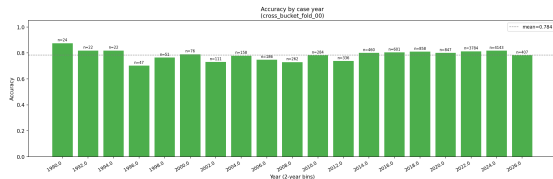
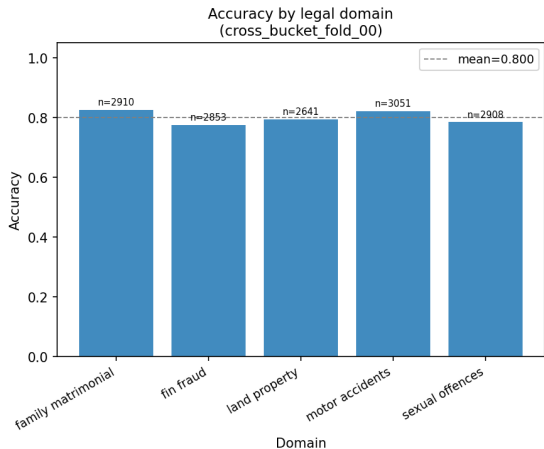
The model is still anchored in the case representation, but it is not purely self-contained. The graph-local legal context changes the prediction materially.

Edge Attention and Gradient Saliency



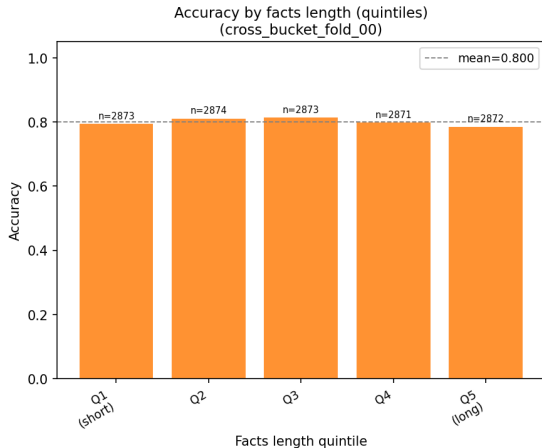
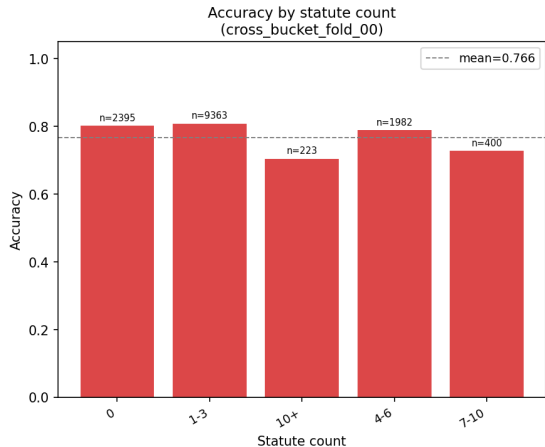
- ▶ reverse edges into the case node dominate attention
- ▶ argument nodes dominate saliency
- ▶ the model operationally hinges on how arguments feed back into the case representation

Error Decomposition: Domain and Time



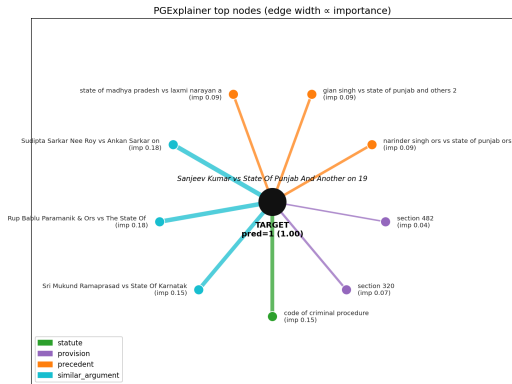
Hard domains remain hard across variants, while temporal drift is limited after very early sparse years are excluded.

Error Decomposition: Structural Complexity



Accuracy falls on **statute-dense cases** and on **the longest factual narratives**. The residual errors are about structural complexity, not about withheld names.

Case-Level Explanation: A Confidently Correct Prediction



Case 10212

Sanjeev Kumar vs State of Punjab (2023)

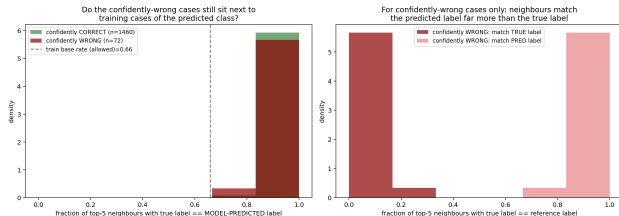
Predicted: **petitioner wins** with probability 0.9998

Ground truth: **petitioner wins**

Evidence the Explainer Extracts

- ▶ governing statute: CrPC
- ▶ top provisions: Section 320 and Section 482 CrPC
- ▶ retrieved neighbours: other Punjab quash-on-compromise wins
- ▶ legal pattern: compromise-driven quashing under inherent High Court power

Case-Level Explanation: When Confidence and Correctness Diverge



Why This Matters

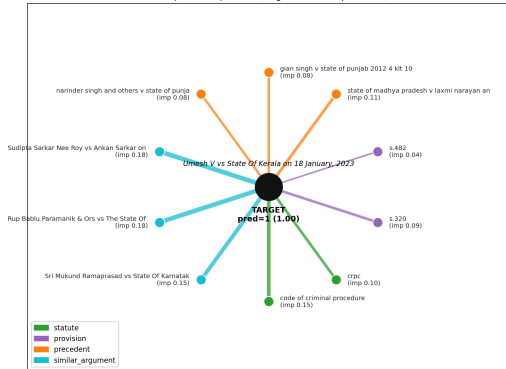
Among high-confidence predictions, **72 cases are confidently wrong**. Their retrieved neighbours overwhelmingly match the *predicted wrong class*, which means the explanation pipeline exposes the failure instead of hiding it.

Structural Signature

A confident error is a case placed inside the wrong outcome neighbourhood in embedding space.

Worked Example of a Confidently Wrong Case

PGExplainer top nodes (edge width $\propto$ importance)



Case 13701

Umesh V vs State of Kerala (2023)

Predicted: **petitioner wins** with probability 0.9998

Ground truth: **petitioner loses**

Interpretation

The local evidence is not random noise. The target case sits in a neighbourhood of structurally similar Kerala win cases, so both prediction and retrieval reinforce the same wrong class.

What the Thesis Establishes

Main Findings

- ▶ The thesis builds a **legally coherent, large-scale Indian legal corpus** from raw judgments with explicit leakage auditing.
- ▶ A reasoning-focused heterogeneous graph improves over strong text-only references on the pooled prediction task.
- ▶ The gain is best described as **structured aggregation of legal context**, not as identity memorization or direct decision leakage.
- ▶ The learned representation is auditable globally and locally.
- ▶ The pooled model transfers partially to an unseen legal domain without fine-tuning.

Most Defensible Claim

Graph-based legal prediction is most credible when it is **structurally faithful, experimentally honest, and auditable case-by-case**.

Data and Evaluation

- ▶ Main numbers are transductive, so entity overlap at test time is partly assumed.
- ▶ Outcome labels are LLM-generated and therefore noisy, especially around procedural dispositions.
- ▶ Leakage filtering lowers risk but cannot mathematically prove zero leakage in every case.

Modeling Limits

- ▶ Ablation margins are often small, so some variants should be treated as a cluster rather than a strict ranking.
- ▶ The pooled optimum is not the universal optimum for every legal domain.
- ▶ Out-of-domain transfer is meaningful but still far below in-domain performance.

Methodological Next Steps

- ▶ inductive temporal and court hold-out splits
- ▶ multi-label or ordinal outcome targets
- ▶ stronger identity-leakage audits
- ▶ connectivity-aware and domain-adaptive modeling

Practical Next Steps

- ▶ expert review of case-level explanations
- ▶ expansion to more legal domains
- ▶ light-weight fine-tuning for unseen domains
- ▶ deployment-time uncertainty signals based on class-boundary proximity, not confidence alone

Closing Takeaway

This thesis argues that legal judgment prediction should be treated as a **structured legal reasoning problem**, not a flat text-classification problem.

The contribution is not just a stronger score, but a pipeline that is **leakage-audited, graph-grounded, interpretable, and partially transferable**.